ELECTRO HYDRAULIC CONTROL SYSTEM

Chapter 3.2

OBJECTIVES

1. Identify the purposes of the Electro Hydraulic Control (EHC) system.

2. Describe the significance of reactor pressure control to BWR operation.

3. Describe how the system accomplishes its purposes.

OBJECTIVES

4. Recognize the purpose of the following limiters:
    a. Load Set
    b. Load Limit
    c. Maximum Combined Flow

OBJECTIVES

5. Recognize the purpose, function and operation of the following EHC system subsystems:
   a. Pressure Control Unit
   b. Speed Control Unit
   c. Desired Load Control Unit
   d. Valve Control Unit
   e. Hydraulic Power Unit

OBJECTIVES

6. Given Figure 3.2-1, describe how the system operates to accomplish the following:
   a. Normal steady state power operations
   b. Power maneuvering
   c. Plant shutdown and cooldown

OBJECTIVES

7. Describe how the EHC system interrelates with the following systems:
   a. Main Steam
   b. Condensate and Feedwater
   c. Reactor Protection
   d. Turbine Building Closed Loop Cooling Water

PURPOSE

The purposes of the EHC System are to:

- provide normal reactor pressure control by controlling steam flow consistent with reactor power
- control reactor pressure during startup, heatup, and cooldown evolutions,
- control the speed and electrical load on the turbine generator,
- provide protection for the main turbine, main generator and main condenser.

A BOILING WATER REACTOR WITHOUT EHC PRESSURE CONTROL

With the reactor operating stable at 100% power:

- A small reduction in main turbine load results in;
- A rise in reactor pressure which results in;
- A reduction in core voids which results in;
- A rise in reactor power which results in;
- A rise in reactor pressure which results in;
- A reduction in core voids which results in;
- A rise in reactor power which results in;
- Etcetera, etcetera, etcetera…

BOILING WATER REACTOR PRESSURE CONTROL

For this reason pressure control in a BWR slaves the turbine to the reactor

To reduce turbine load, you must FIRST reduce reactor power which results in;

- A reduction in reactor steam pressure which results in;
- Lower steam pressure sensed at the turbine which results in;
- The EHC regulator closing the turbine control valves to restore pressure which results in;
- A turbine load reduction and stabilized pressure.

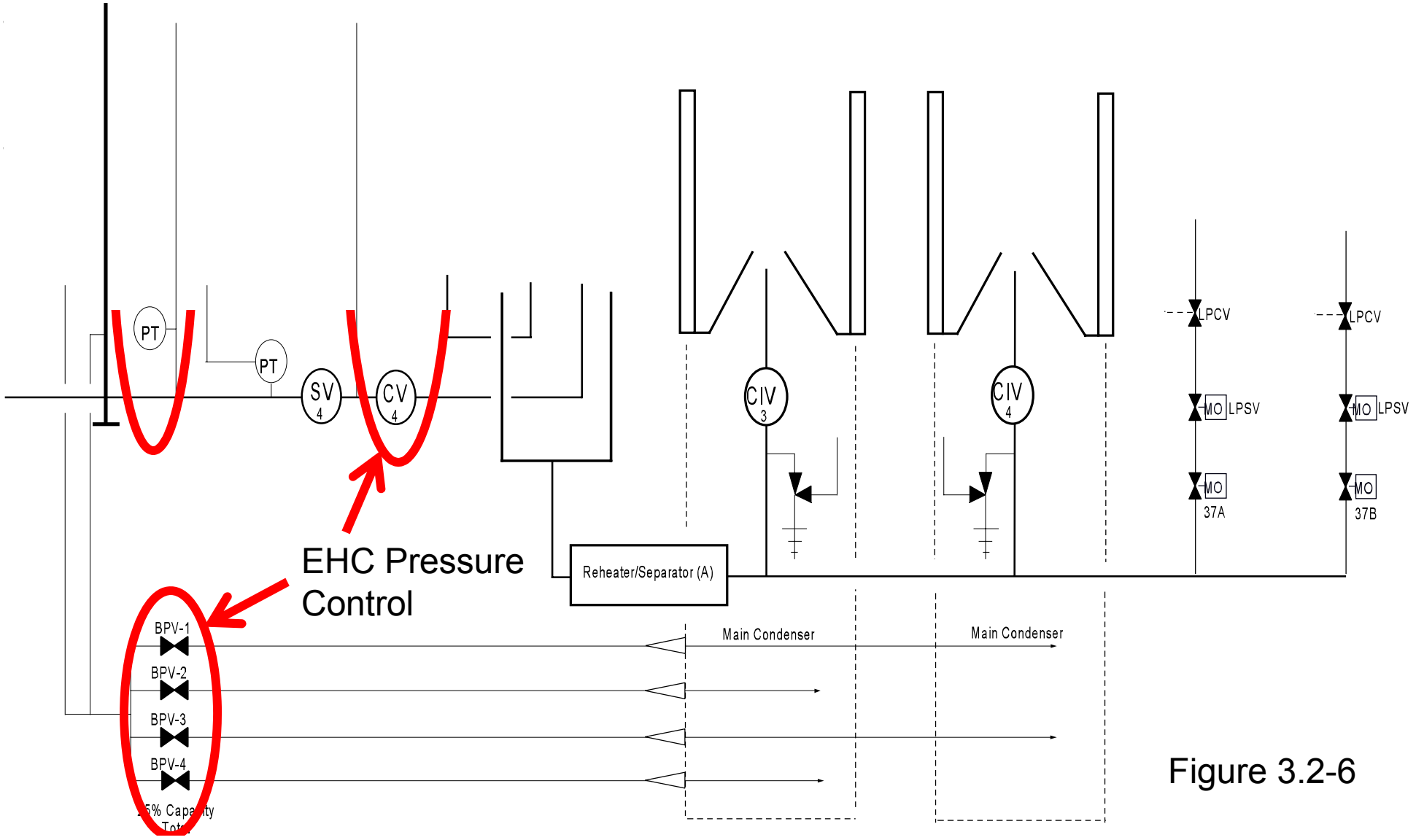

PT
PT
SV
4
CV
4
CIV
3
CIV
4
LPCV
MO
LPSV
MO
37A
LPCV
MO
LPSV
MO
37B
EHC Pressure
Control
Reheater/Separator (A)
Main Condenser
Main Condenser
BPV-1
BPV-2
BPV-3
BPV-4

Figure 3.2-6

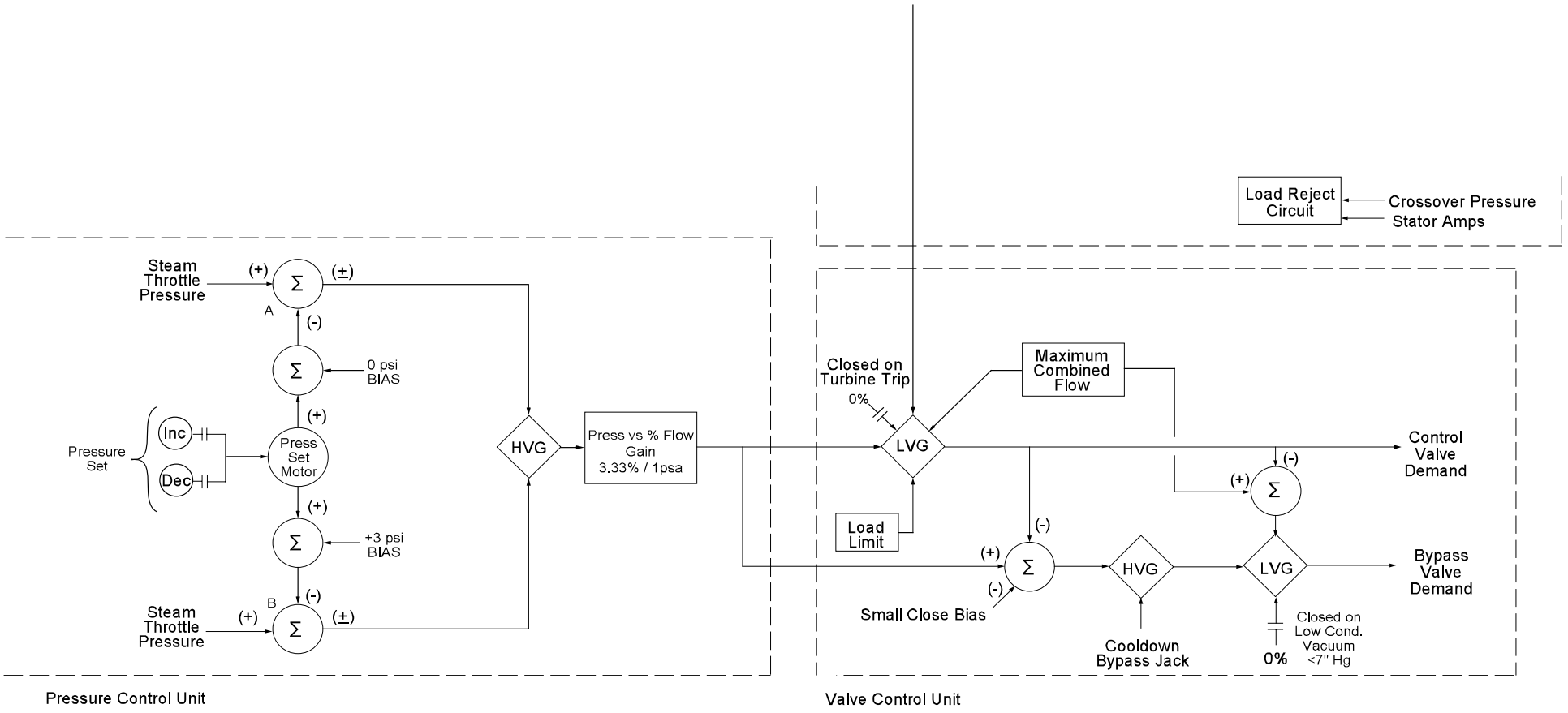

Steam Throttle Pressure
(+)
Σ
(±)
A
(-)
Σ
0 psi BIAS
(+)
Pressure Set
Inc
Dec
Press Set Motor
(+)
Σ
+3 psi BIAS
(-)
B
Steam Throttle Pressure
(+)
Σ
(±)
HVG
Press vs % Flow Gain 3.33% / 1psa
Pressure Control Unit
Load Reject Circuit
Crossover Pressure
Stator Amps
Closed on Turbine Trip
0%
Maximum Combined Flow
LVG
Control Valve Demand
(-)
(+)
Σ
Load Limit
(-)
(+)
Σ
(-)
Small Close Bias
HVG
Cooldown Bypass Jack
LVG
Bypass Valve Demand
Closed on Low Cond. Vacuum <7" Hg
0%
Valve Control Unit

Figure 3.2-1

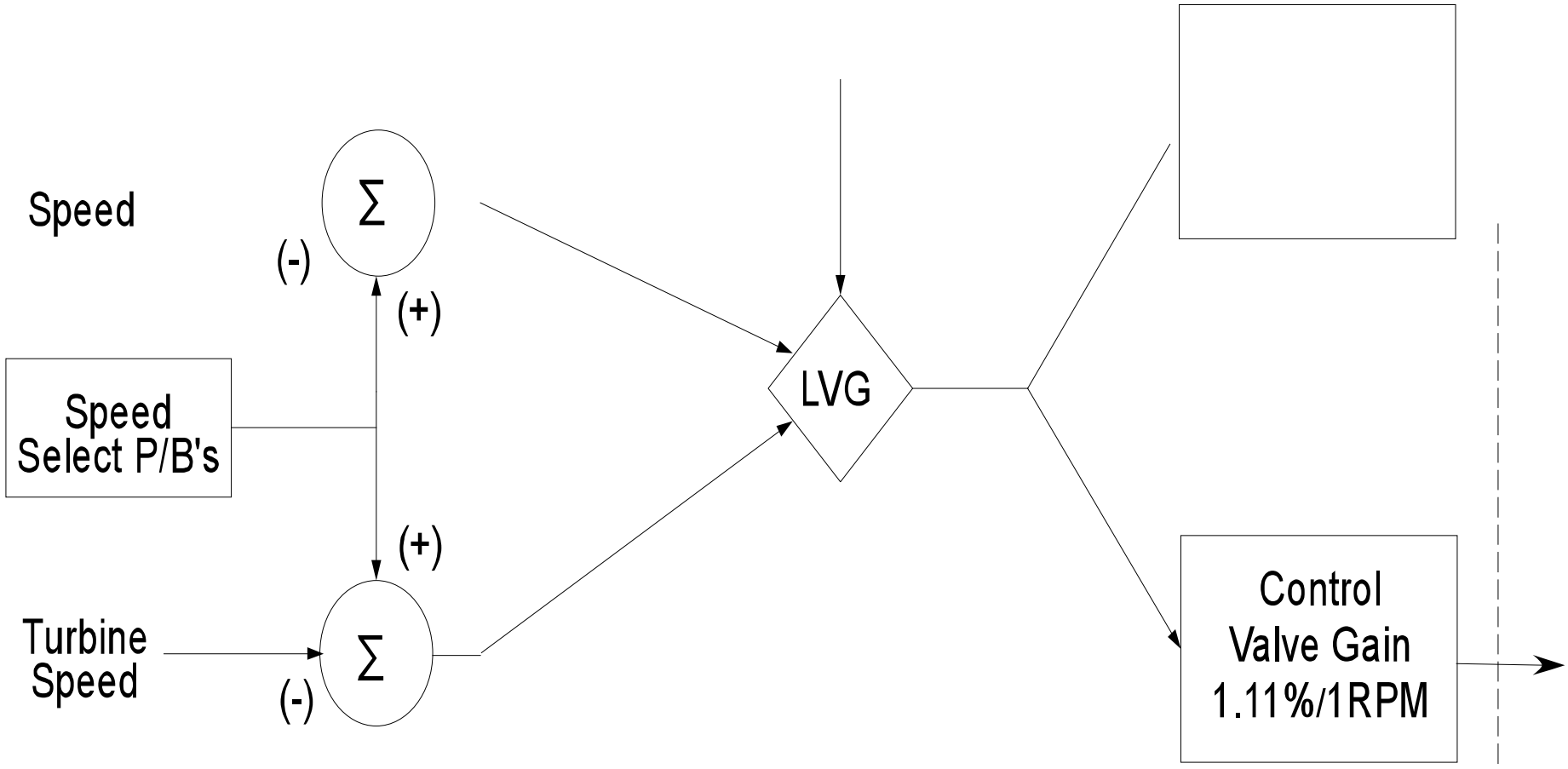

Speed
Σ
(-)
(+)
Speed
Select P/B's
(+)
Turbine
Speed
Σ
(-)
LVG
Control
Valve Gain
1.11%/1RPM

Figure 3.2-1c

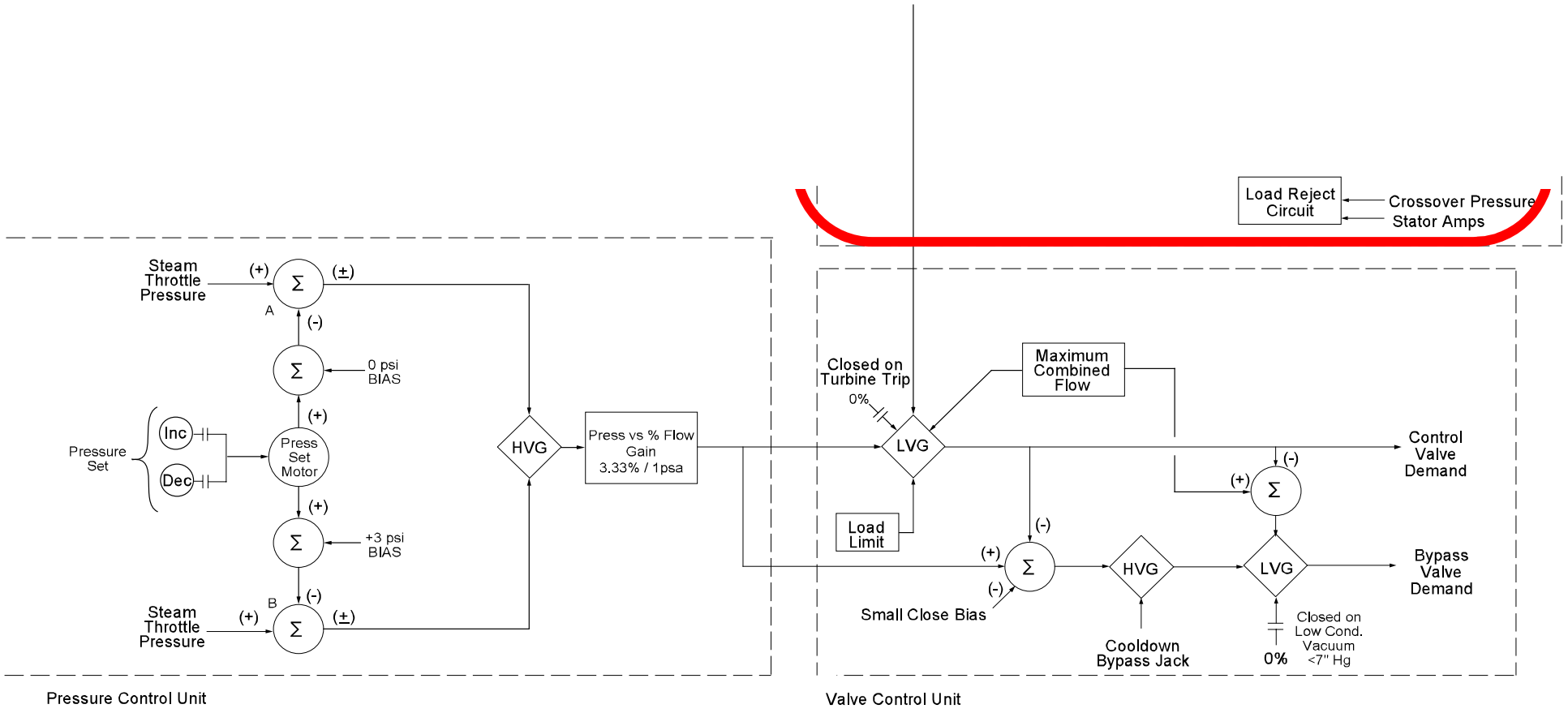

Load Reject Circuit
Crossover Pressure
Stator Amps
Steam Throttle Pressure
(+)
Σ
(±)
A
(-)
Σ
0 psi BIAS
(+)
Pressure Set
Inc
Dec
Press Set Motor
(+)
Σ
+3 psi BIAS
(-)
B
Steam Throttle Pressure
(+)
Σ
(±)
HVG
Press vs % Flow Gain 3.33% / 1psa
Pressure Control Unit
Closed on Turbine Trip
0%
Maximum Combined Flow
LVG
Control Valve Demand
(-)
(+)
Σ
Load Limit
(-)
(+)
Σ
(-)
Small Close Bias
HVG
Cooldown Bypass Jack
LVG
Bypass Valve Demand
Closed on Low Cond. Vacuum <7" Hg
0%
Valve Control Unit

Figure 3.2-1

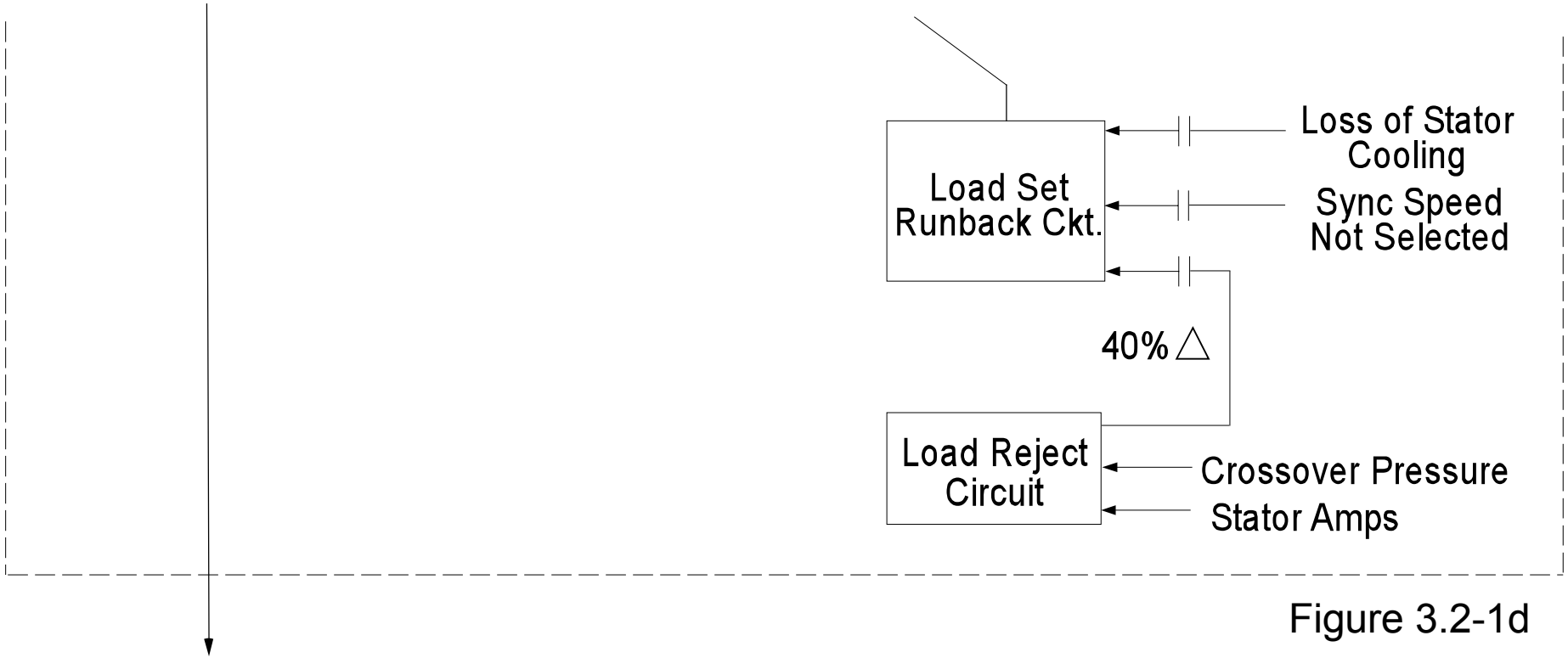

Load Set
Runback Ckt.
Loss of Stator
Cooling
Sync Speed
Not Selected
40% △
Load Reject
Circuit
Crossover Pressure
Stator Amps

Figure 3.2-1d

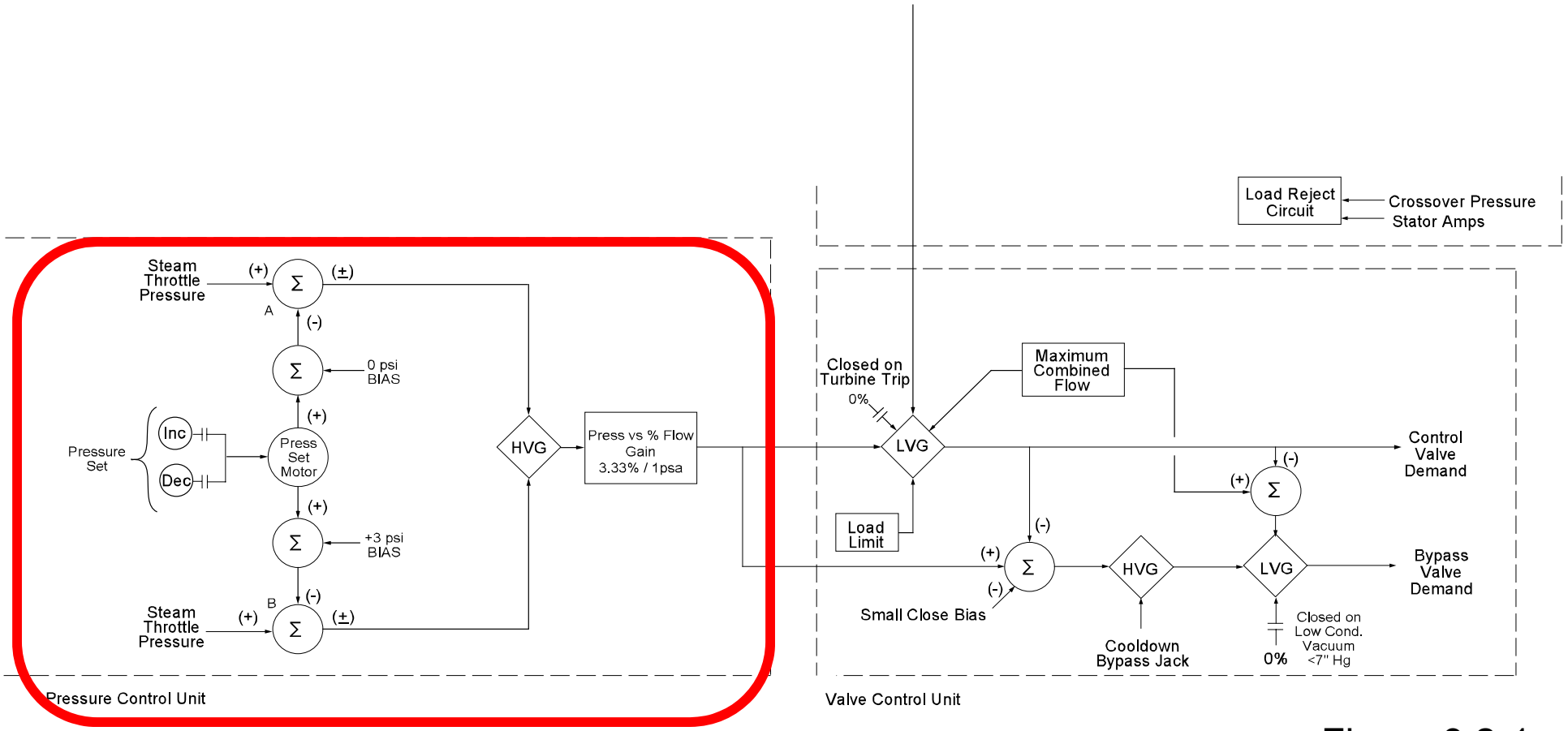
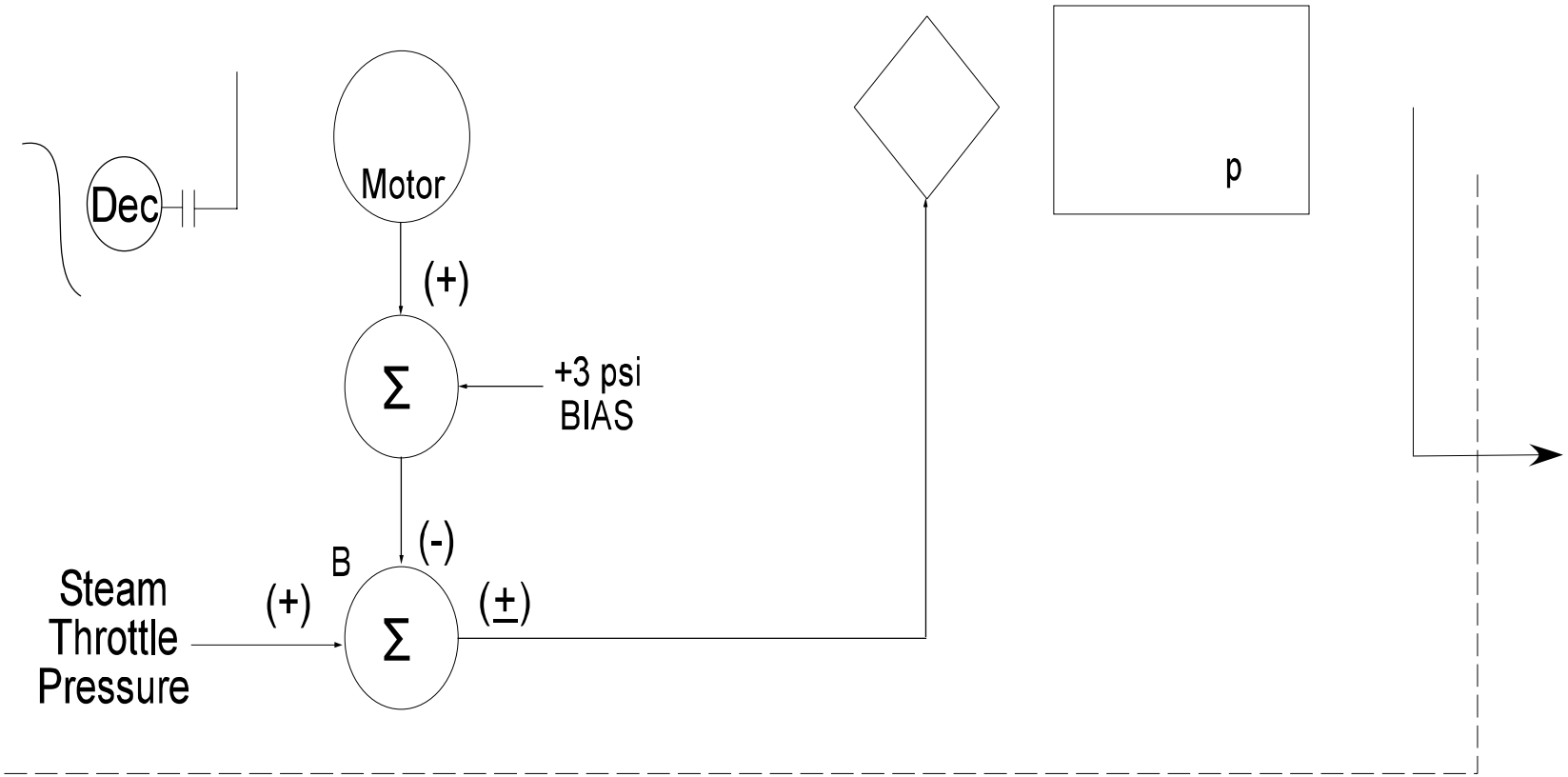

Steam Throttle Pressure
(+)
Σ
(±)
A
(-)
Σ
0 psi BIAS
(+)
Inc
Pressure Set
Dec
Press Set Motor
(+)
Σ
+3 psi BIAS
(-)
B
Steam Throttle Pressure
(+)
Σ
(±)
HVG
Press vs % Flow Gain 3.33% / 1psa
Pressure Control Unit
Load Reject Circuit
Crossover Pressure
Stator Amps
Closed on Turbine Trip
0%
Maximum Combined Flow
LVG
Control Valve Demand
(-)
(+)
Σ
Load Limit
(-)
(+)
Σ
(-)
Small Close Bias
HVG
Cooldown Bypass Jack
LVG
Bypass Valve Demand
Closed on Low Cond. Vacuum <7" Hg
0%
Valve Control Unit

Figure 3.2-1



Pressure Control Unit

Figure 3.2-1a

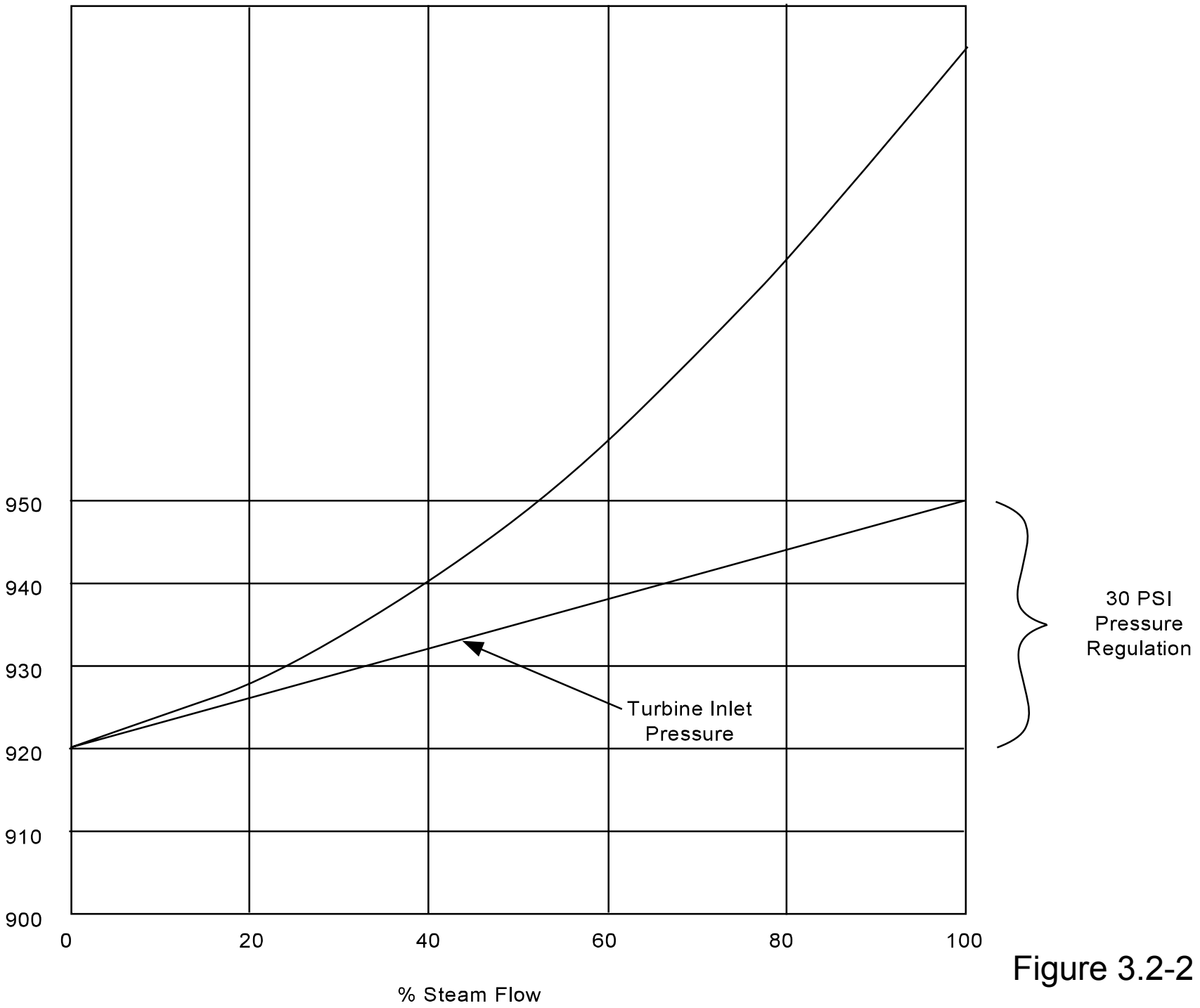

950
940
930
920
910
900
0
20
40
60
80
100
% Steam Flow
Turbine Inlet
Pressure
30 PSI
Pressure
Regulation

Figure 3.2-2

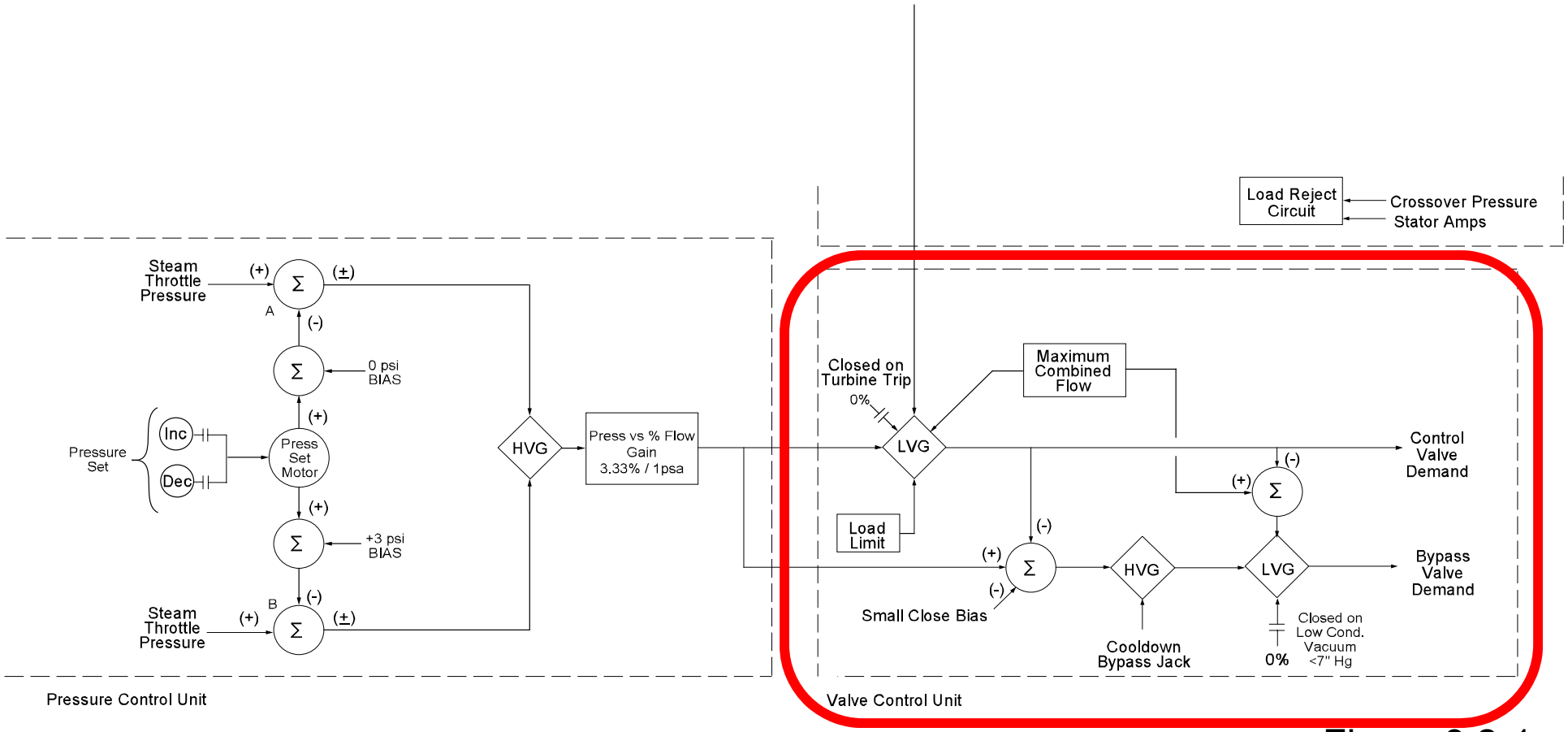

Steam Throttle Pressure
(+)
Σ
(±)
A
(-)
Σ
0 psi BIAS
(+)
Pressure Set
Inc
Dec
Press Set Motor
(+)
Σ
+3 psi BIAS
(-)
B
Steam Throttle Pressure
(+)
Σ
(±)
HVG
Press vs % Flow Gain 3.33% / 1psa
Pressure Control Unit
Load Reject Circuit
Crossover Pressure
Stator Amps
Closed on Turbine Trip
0%
Maximum Combined Flow
LVG
Control Valve Demand
(-)
(+)
Σ
Load Limit
(-)
(+)
Σ
(-)
Small Close Bias
HVG
Cooldown Bypass Jack
LVG
Bypass Valve Demand
Closed on Low Cond. Vacuum <7" Hg
0%
Valve Control Unit

Figure 3.2-1

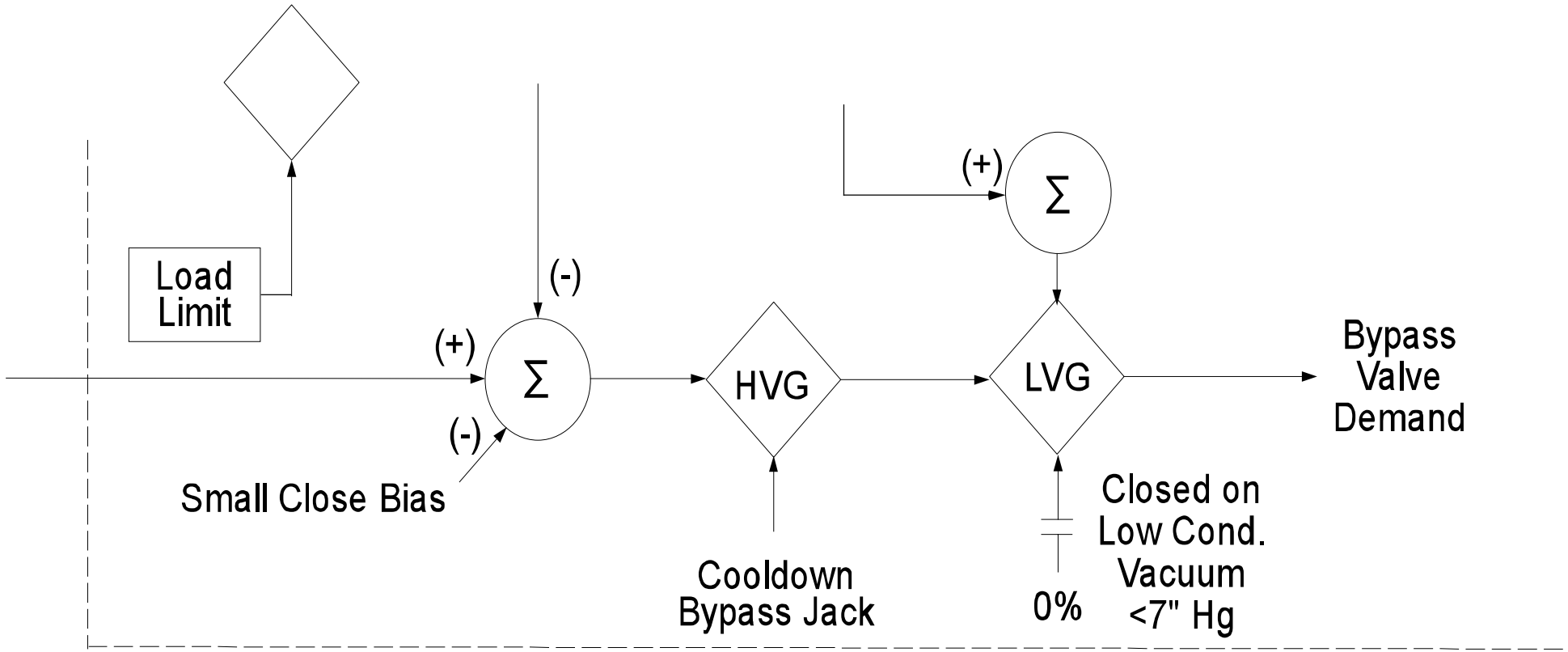

Load
Limit
(+)
(-)
Σ
(-)
Small Close Bias
HVG
Cooldown
Bypass Jack
(+)
Σ
LVG
0%
Closed on
Low Cond.
Vacuum
<7" Hg
Bypass
Valve
Demand
Valve Control Unit

Figure 3.2-1b

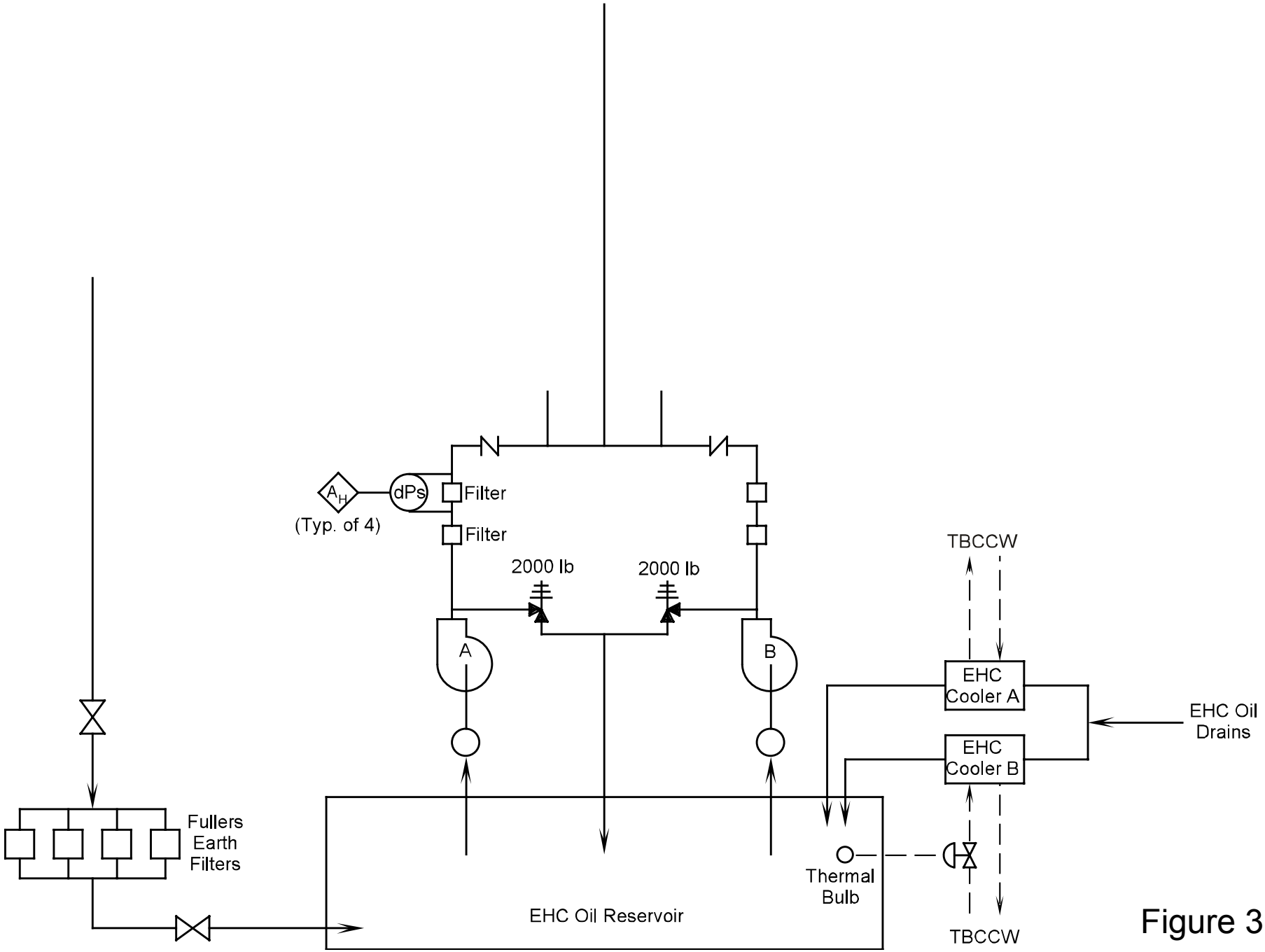

A_H
dPs
(Typ. of 4)
Filter
Filter
2000 lb
2000 lb
A
B
TBCCW
EHC
Cooler A
EHC
Cooler B
EHC Oil
Drains
Fullers
Earth
Filters
Thermal
Bulb
EHC Oil Reservoir
TBCCW

Figure 3.2-4

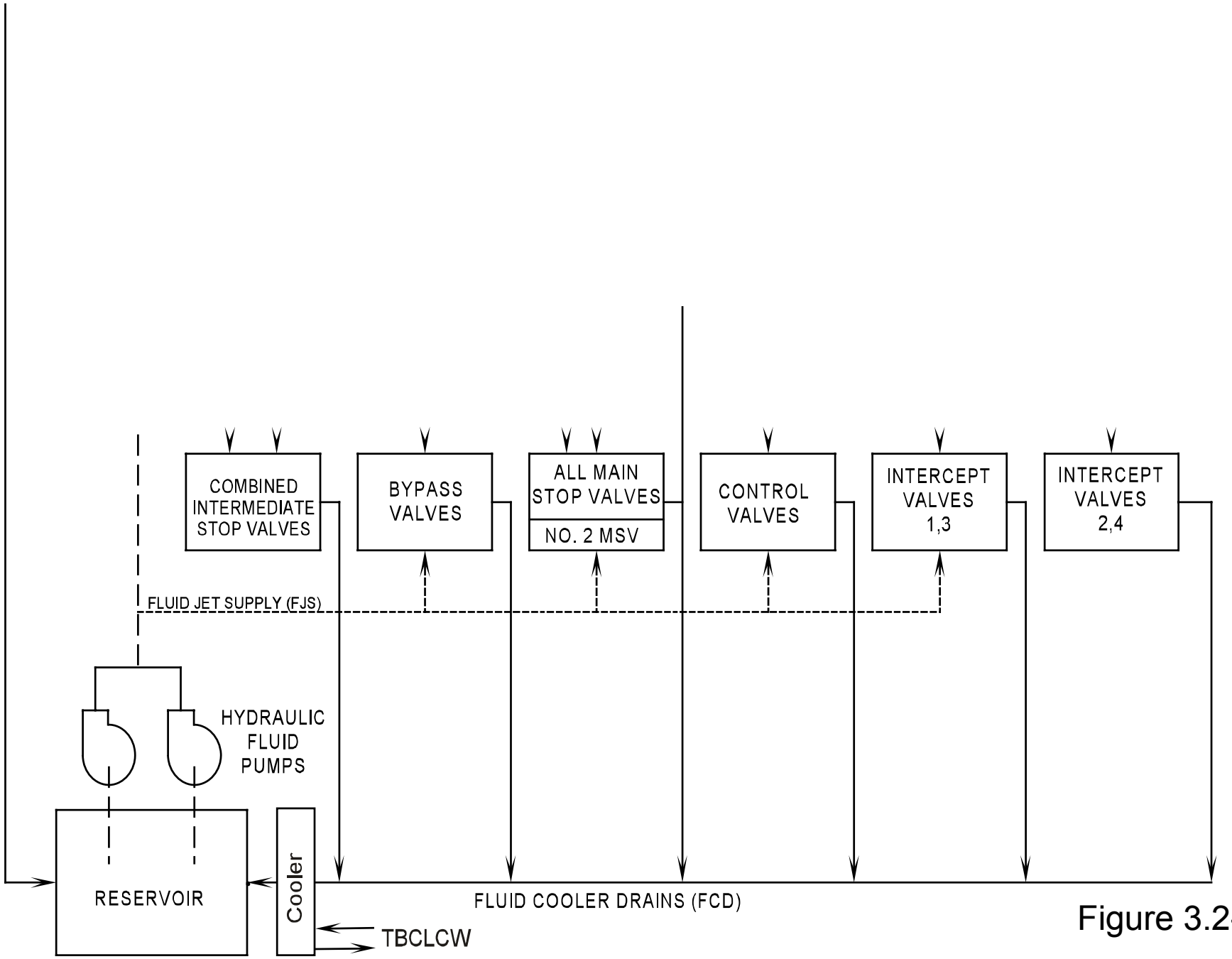
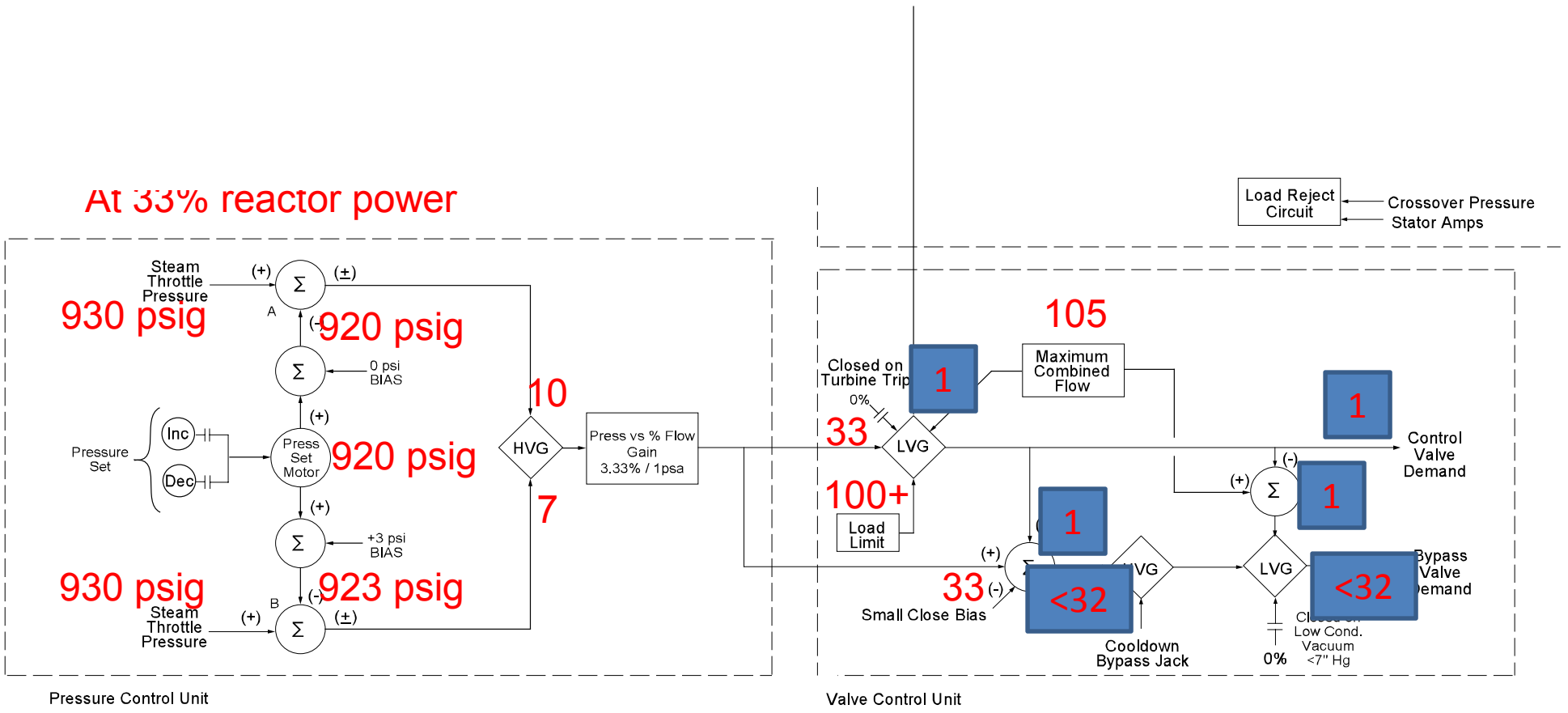

COMBINED INTERMEDIATE STOP VALVES
BYPASS VALVES
ALL MAIN STOP VALVES
NO. 2 MSV
CONTROL VALVES
INTERCEPT VALVES 1,3
INTERCEPT VALVES 2,4
FLUID JET SUPPLY (FJS)
HYDRAULIC FLUID PUMPS
RESERVOIR
Cooler
TBCLCW
FLUID COOLER DRAINS (FCD)

Figure 3.2-5

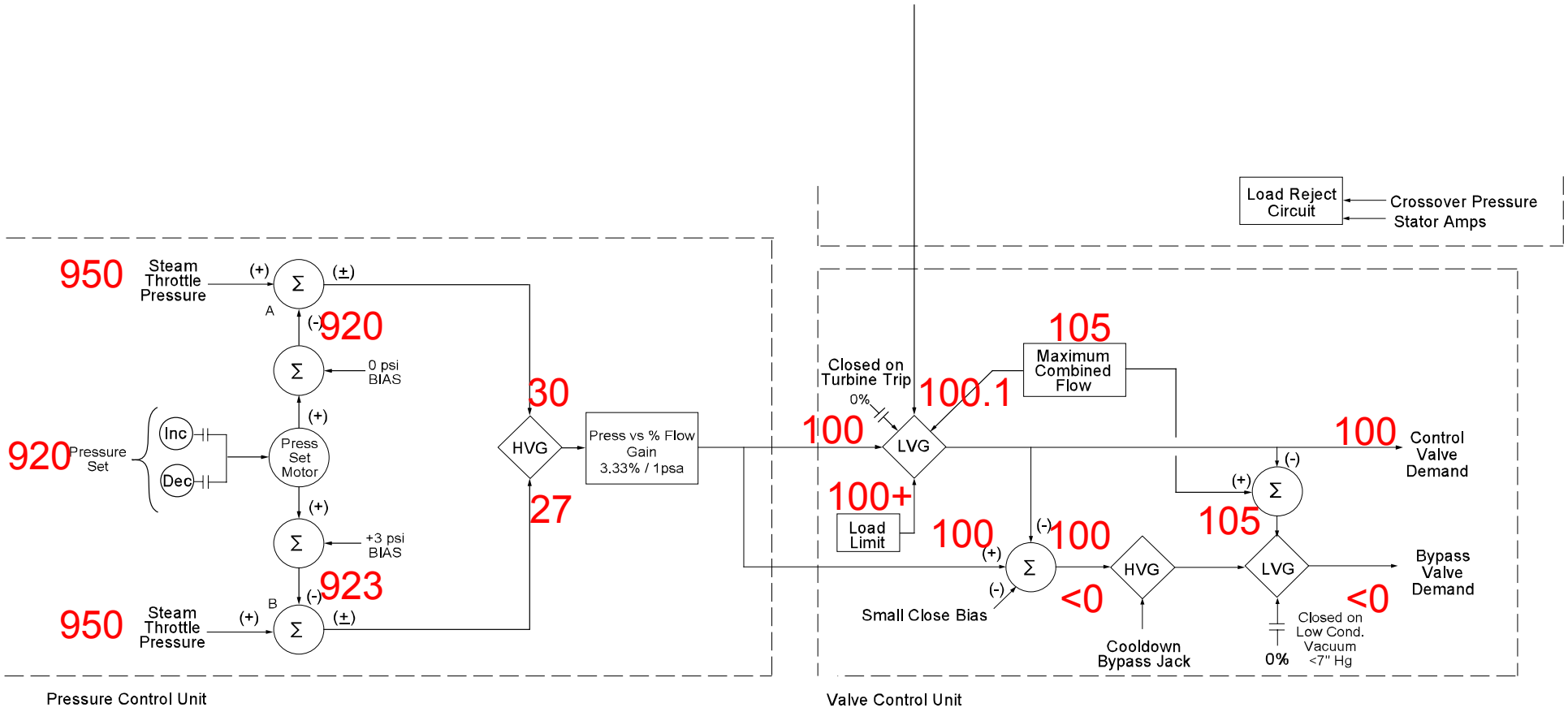
TABLE 3.2-1 Turbine Trip Conditions

Trip	Setpoint and Reason
Reactor Vessel High Level	56.5" An excessively high water level could result in moisture carry over to the turbine resulting in blade erosion or damage.
EHC Fluid Header Pressure Low	<1100 psig Loss of EHC oil pressure would indicate a potential for no control since this is the hydraulic source for valve actuation.
Thrust Bearing Wear	~35 mils Indicates a potential misalignment between the diaphragms and buckets which could result in mechanical damage to the turbine.
Mechanical Overspeed Trip Backup Electrical Overspeed Trip	110/112% Indicates potential turbine damage due to excessive turbine speed and the resultant forces and misalignment.
Exhaust Hood High Temperature	225ºF Excessive temperatures would result in thermal stress, damage to exhaust hood or potential misalignment.
Stator Cooling Failure	<13 psig or >95 ºC Indicates operation of the generator under abnormal conditions. To prevent damage to the generator the turbine is tripped after a 70 second time delay if generator amps are >5811.
Low Main Shaft Oil Pump Pressure	<105 psig @ >1300 RPM A turbine trip is required to prevent bearing damage due to a loss of lubricating oil if turbine speed is >1300 RPM.
Low Bearing Oil Pressure	8 psig Prevent turbine damage due to loss of lubrication
Loss of Both Speed Feedback Channels to EHC Unit	Trips turbine because of potential overspeed condition if turbine speed is >200 RPM
Low Condenser Vacuum	22.5" Hg Indicates a loss of heat sink and operation of turbine at conditions for which it was not designed
Turbine/Generator High Vibration	10 mils Anticipates turbine damage due to excessive vibration

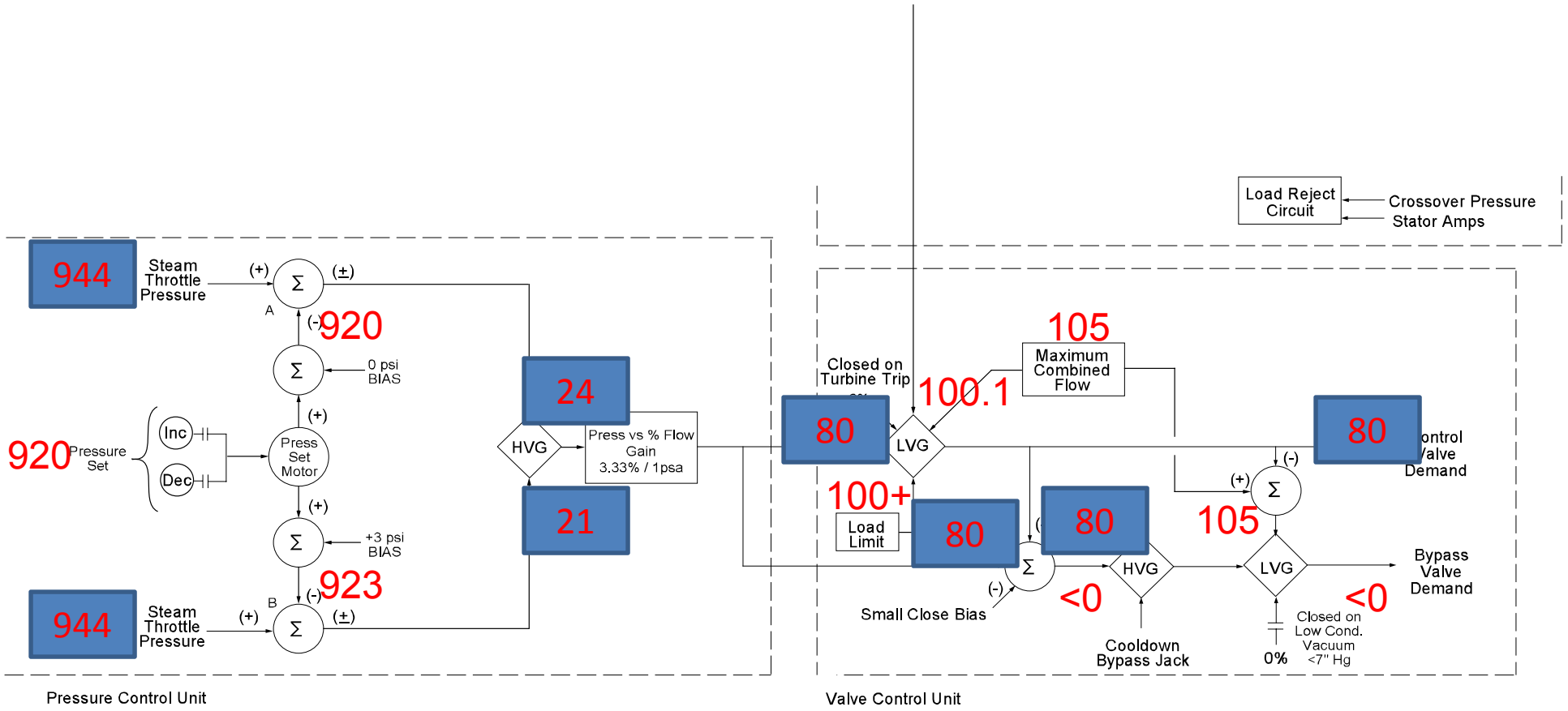


INITIAL TURBINE ROLL TO 1800 RPM

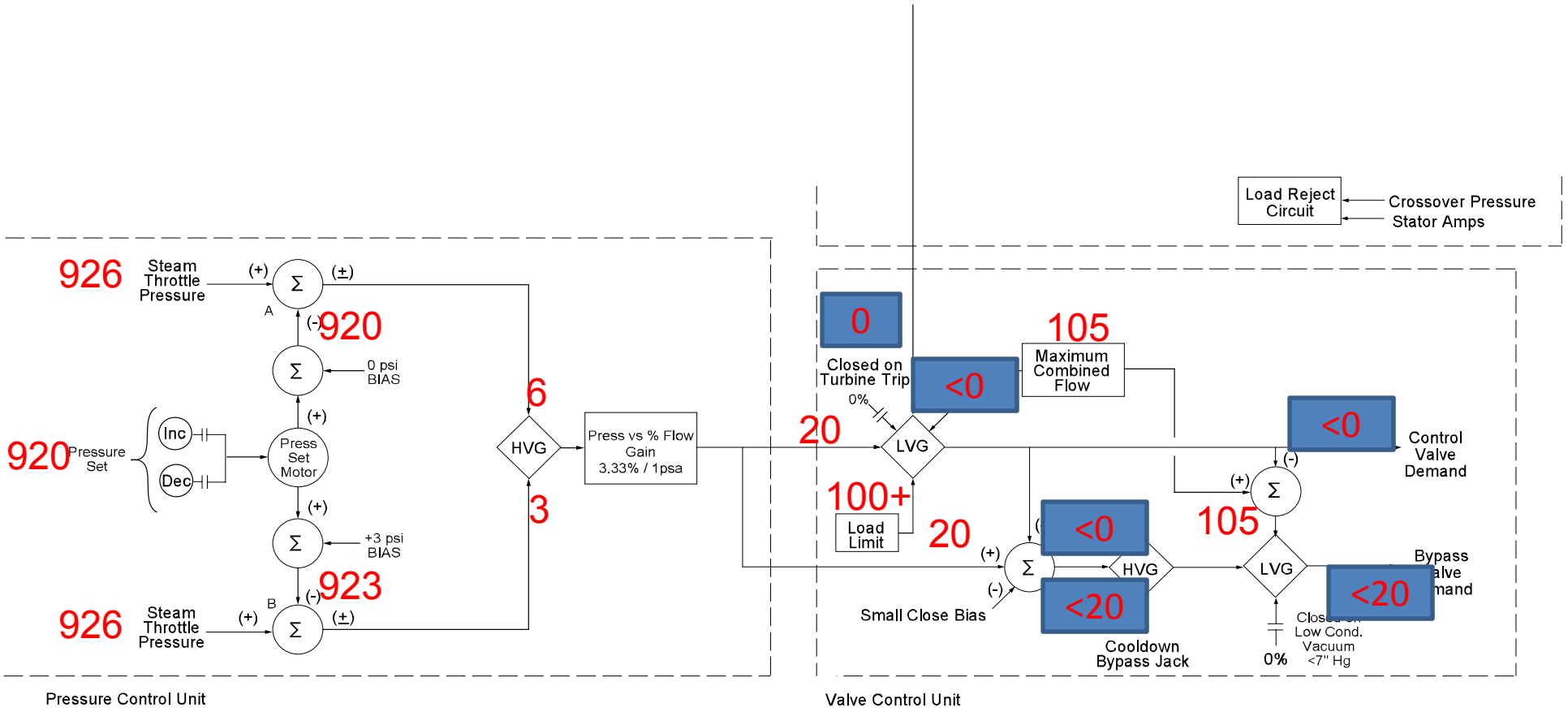
Figure 3.2-1



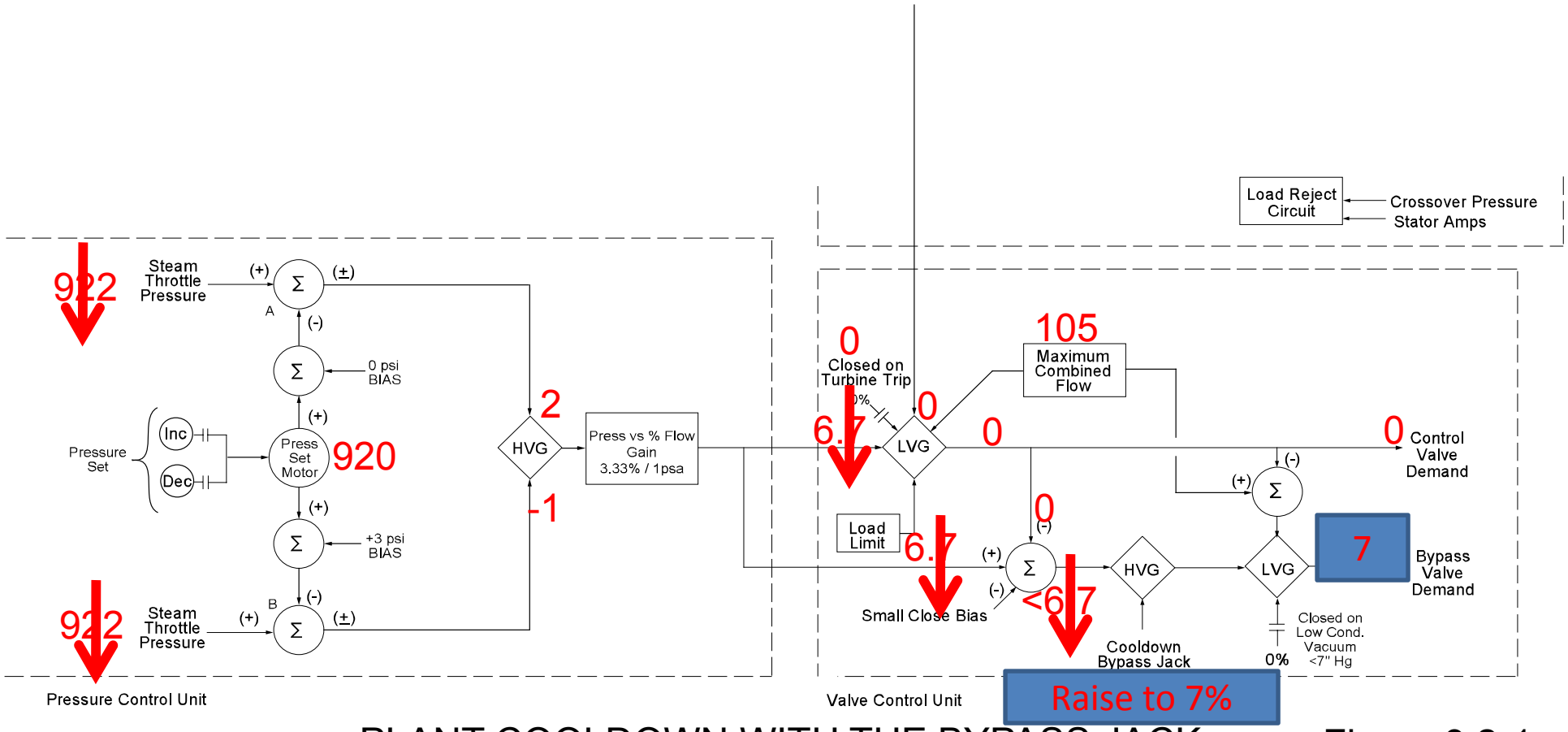
NORMAL STEADY STATE FULL POWER OPERATION Figure 3.2-1



REACTOR POWER REDUCTION FROM 100% TO 80% Figure 3.2-1

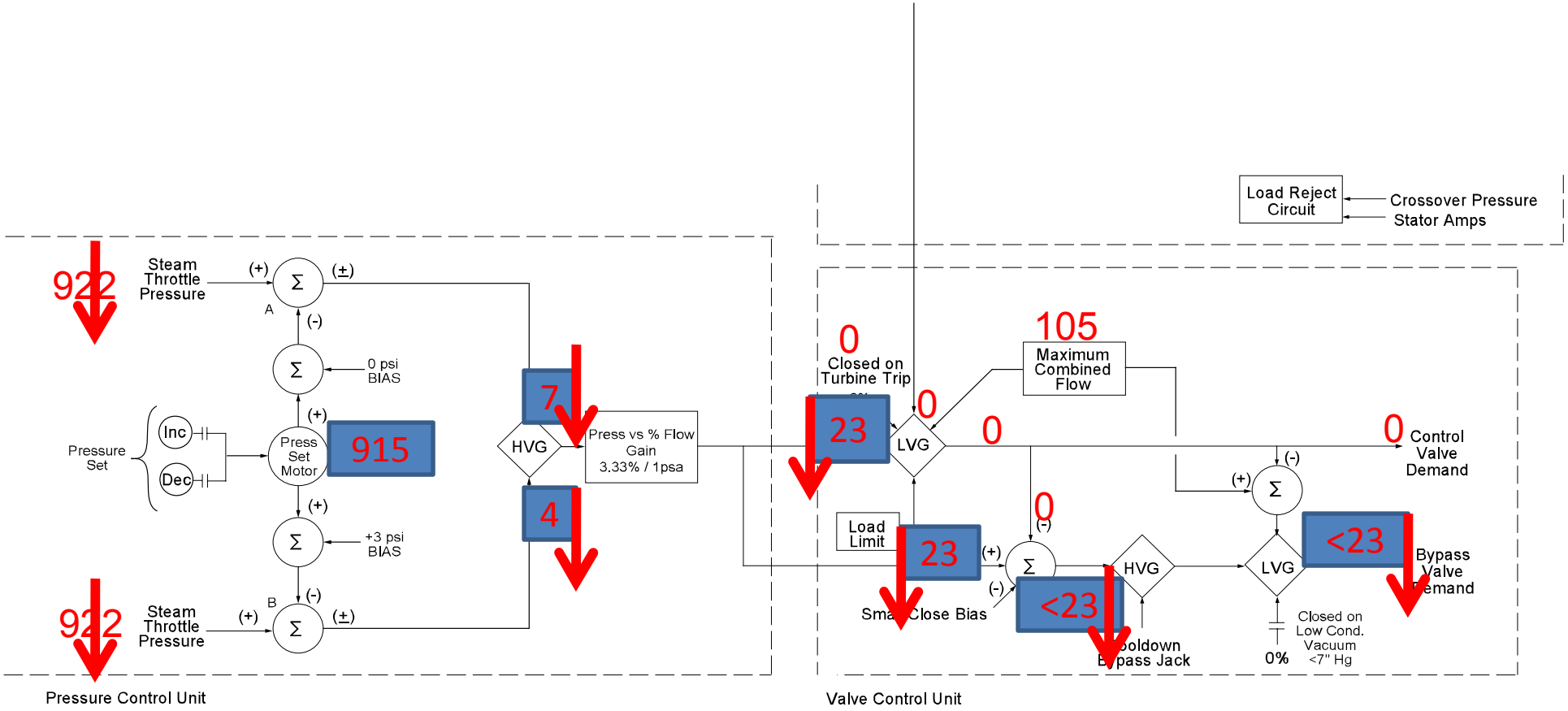


TURBINE SHUTDOWN FROM 20% REACTOR POWER Figure 3.2-1



PLANT COOLDOWN WITH THE BYPASS JACK

Figure 3.2-1



PLANT COOLDOWN WITH PRESSURE SET

Figure 3.2-1

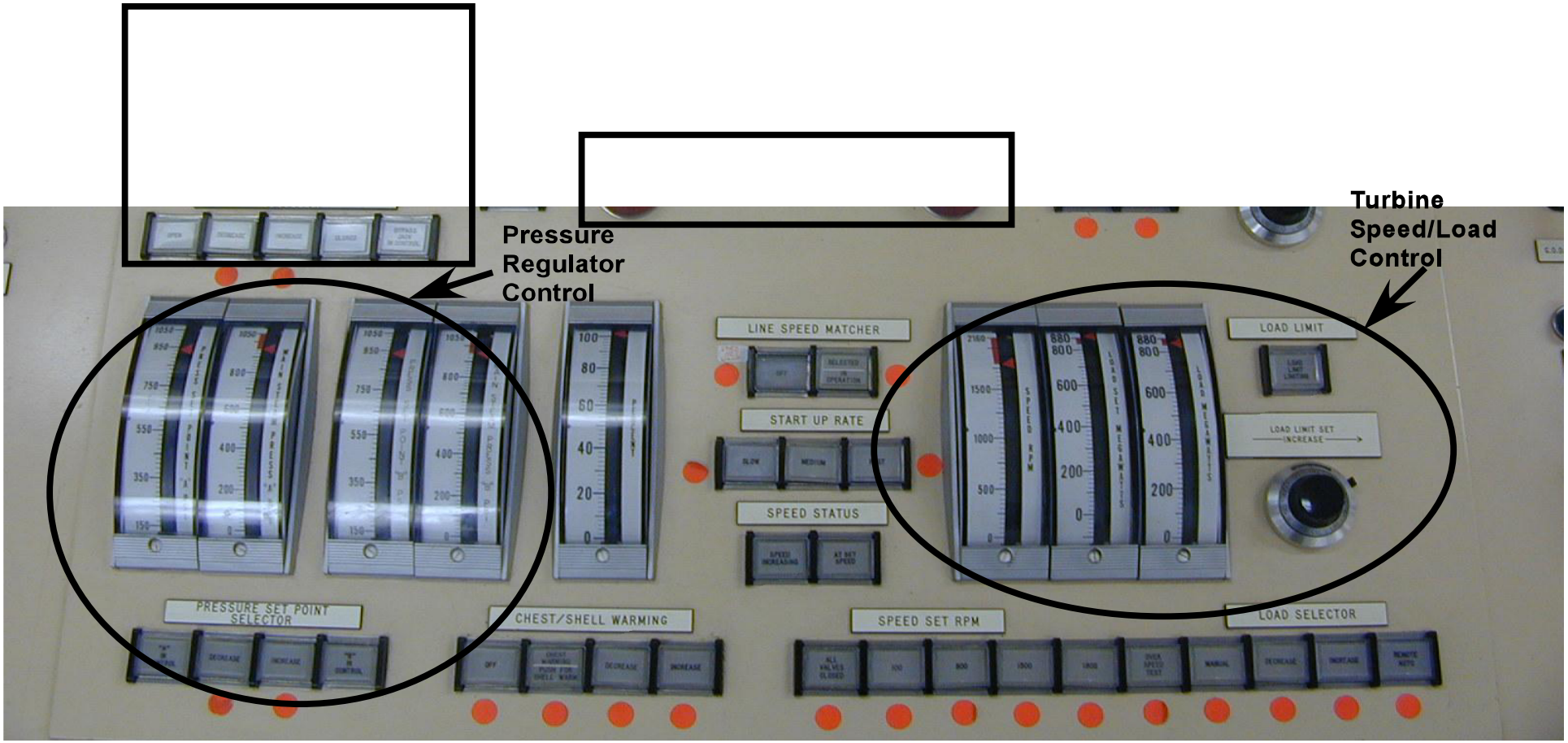

Pressure
Regulator
Control
Turbine
Speed/Load
Control
LINE SPEED MATCHER
START UP RATE
SPEED STATUS
LOAD LIMIT
LOAD LIMIT SET
INCREASE
PRESSURE SET POINT
SELECTOR
CHEST/SHELL WARMING
SPEED SET RPM
LOAD SELECTOR

Figure 3.2-3

SYSTEM INTERFACES

- Main Steam
  - EHC controls TSV, TCV, BPV, ISV, ICV positions
  - Main Steam provides EHC sensed pressure input
- Condensate/Feedwater
  - EHC causes extraction non-return valve closure on a turbine trip.
  - BPV's are interlocked closed at less than 7 inches mercury vacuum.
- Reactor Protection
  - RPS receives scram signals for TSV Closure and TCV Fast Closure from EHC
- Turbine Building Closed Loop Cooling
  - TBCLC cools the EHC hydraulic oil.

OBJECTIVES

1. Identify the purposes of the Electro Hydraulic Control (EHC) system.

2. Describe the significance of reactor pressure control to BWR operation.

3. Describe how the system accomplishes its purposes.

OBJECTIVES

4. Recognize the purpose of the following limiters:
   - Load Set
   - Load Limit
   - Maximum Combined Flow

OBJECTIVES

5. Recognize the purpose, function and operation of the following EHC system subsystems:
   - Pressure Control Unit
   - Speed Control Unit
   - Desired Load Control Unit
   - Valve Control Unit
   - Hydraulic Power Unit

OBJECTIVES

6. Given Figure 3.2-1, describe how the system operates to accomplish the following:
   - Normal steady state power operations
   - Power maneuvering
   - Plant shutdown and cooldown

OBJECTIVES

7. Describe how the EHC system interrelates with the following systems:
   - Main Steam
   - Condensate and Feedwater
   - Reactor Protection
   - Turbine Building Closed Loop Cooling Water